

Calc 1 – Applications of the Derivative

Worksheet A

1. Find all critical points of $f(x) = x^3 - 3x^2 - 9x + 1$ and classify them.
2. Find the absolute extrema of $f(x) = x^3 - 12x$ on $[-3, 3]$.
3. A rectangle has perimeter 20. Find the dimensions that maximize its area.
4. Verify MVT for $f(x) = x^2$ on $[1, 3]$ and find the c guaranteed.
5. A ladder 10 ft long leans against a wall. Its base slides out at 2 ft/s. How fast is the top sliding down when the base is 6 ft from the wall?

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Worksheet B

1. Find the local extrema of $f(x) = \frac{x}{x^2 + 1}$.
2. Find the absolute extrema of $f(x) = x + 2 \sin x$ on $[0, 2\pi]$.
3. A box with a square base and open top has volume 32,000 cm³. Find the dimensions minimizing surface area.
4. Use linear approximation to estimate $\sqrt[3]{8.1}$.
5. Use one step of Newton's method on $f(x) = x^2 - 5$ starting from $x_0 = 2$ to estimate $\sqrt{5}$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $f' = 3x^2 - 6x - 9 = 3(x - 3)(x + 1) = 0$ at $x = -1, 3$. $f'' = 6x - 6$: $f''(-1) = -12 < 0$ (**local max**, value 6); $f''(3) = 12 > 0$ (**local min**, value -26).
2. $f'(x) = 3x^2 - 12 = 0$ at $x = \pm 2$. Values: $f(-3) = -27 + 36 = 9$, $f(-2) = 16$, $f(2) = -16$, $f(3) = -9$. **Max** = 16 **at** $x = -2$; **min** = -16 **at** $x = 2$.
3. Let sides be $x, 10 - x$. Area $A = x(10 - x)$, $A' = 10 - 2x = 0 \Rightarrow x = 5$. So $\boxed{5 \times 5}$ (a square), area 25.
4. f is continuous on $[1, 3]$, differentiable on $(1, 3)$. Slope: $\frac{9-1}{2} = 4 = f'(c) = 2c \Rightarrow \boxed{c = 2}$.
5. $x^2 + y^2 = 100$, so $2xx' + 2yy' = 0$. At $x = 6$: $y = 8$, $x' = 2$. Then $y' = -xx'/y = -12/8 = \boxed{-1.5}$ ft/s (top descends at 1.5 ft/s).

Worksheet B.

1. $f' = \frac{(1)(x^2+1)-x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$. Zero at $x = \pm 1$. Sign change $+$ $\rightarrow$ $-$ at $x = 1$ (**max**, $f(1) = 1/2$); $- \rightarrow +$ at $x = -1$ (**min**, $f(-1) = -1/2$).
2. $f'(x) = 1 + 2\cos x = 0 \Rightarrow \cos x = -1/2$, so $x = 2\pi/3, 4\pi/3$ in $[0, 2\pi]$. Values: $f(0) = 0$, $f(2\pi/3) = 2\pi/3 + \sqrt{3}$, $f(4\pi/3) = 4\pi/3 - \sqrt{3}$, $f(2\pi) = 2\pi$. **Max** = 2π **at** $x = 2\pi$; **min** = 0 **at** $x = 0$.
3. Let base = x , height = h . Volume $x^2h = 32000 \Rightarrow h = 32000/x^2$. Surface area $S = x^2 + 4xh = x^2 + 128000/x$. $S' = 2x - 128000/x^2 = 0 \Rightarrow x^3 = 64000 \Rightarrow x = 40$, $h = 20$. **Dimensions** $40 \times 40 \times 20$ **cm**.
4. Near $a = 8$: $f(x) = x^{1/3}$, $f'(x) = \frac{1}{3}x^{-2/3}$, $f'(8) = 1/12$. $\sqrt[3]{8.1} \approx 2 + \frac{0.1}{12} \approx \boxed{2.00833}$.
5. $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{-1}{4} = \boxed{2.25}$ (true $\sqrt{5} \approx 2.2361$).